Woohyeok Kim

Integrated M.S./Ph.D. Student
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EDUCATION

Sep. 2022 ~ Present	**Pohang University of Science and Technology** Graduate School of Artificial Intelligence *Advisor: Sunghyun Cho* *Ph.D. Student* GPA: 4.11 / 4.3	Pohang, Korea
Mar. 2017 ~ Aug. 2022	**Sejong University** Unmanned Vehicle Engineering *B.S in Intelligent Mechatronics Engineering, Summa Cum Laude (1/84)* GPA: 4.43 / 4.5	Seoul, Korea

RESEARCH INTERESTS

- Computer Vision, Computational Photography, Camera ISP, Image Restoration & Enhancement, Burst Imaging, HDR Imaging, Optimal Imaging Systems

PUBLICATIONS (INTERNATIONAL)

1. Eunsue Choi*, Gyeongtae Kim*, Eunji Lee*, Seungwoo Yoon, Youngmo Jeong, **Woohyeok Kim**, Minseok Choi, Sunghyun Cho, Junsuk Rho, Seung-Hwan Baek, "Learned π Metalens for Wide-field-of-view Near-infrared Imaging", ACM Transactions on Graphics (SIGGRAPH Asia 2026)
2. Yunkyu Lee, **Woohyeok Kim**, Sunghyun Cho, "Realistic Compound-Lens Defocus Blur Synthesis", ECCV 2026
3. Jinnyeong Kim, Juhyung Choi, **Woohyeok Kim**, Sunghyun Cho, Seung-Hwan Baek, "Depth-guided Multi-view Exposure Bracketing for HDR Robot Vision", ECCV 2026
4. **Woohyeok Kim**, Jaesung Rim, Daeyeon Kim, Sunghyun Cho, "Dynamic Exposure Burst Image Restoration", CVPR 2026
5. Jaesung Rim, **Woohyeok Kim**, Haeyun Lee, Heemin Yang, Ke Wang, Sunghyun Cho, "Gyro-based Deep Video Deblurring", CVPR 2026
6. Jiyun Won, Heemin Yang, **Woohyeok Kim**, Jungseul Ok, Sunghyun Cho, "POS-ISP: Pipeline Optimization at the Sequence Level for Task-aware ISP", CVPR 2026 Findings
7. **Woohyeok Kim***, Genu Kim*, Junyong Lee, Seungyong Lee, Sunghyun Cho, "ParamISP: Learned

Forward and Inverse ISPs using Camera Parameters", CVPR 2024

8. Sanghyun Kim*, Minjung Lee*, **Woohyeok Kim**, Deunsol Jung, Jaesung Rim, Sunghyun Cho, Minsu Cho, "Burst Image Super-Resolution with Base Frame Selection", CVPR 2024 Workshops

PUBLICATIONS (DOMESTIC)

1. Jiyun Won, Heemin Yang, **Woohyeok Kim**, Jungseul Ok, Sunghyun Cho, "Sequence-Level Pipeline Optimization for Task-Aware ISP", IPIU 2026
2. Yunkyu Lee, **Woohyeok Kim**, Sunghyun Cho, "Lens Modeling for Integrated Simulation of Optical Image Degradation", KCGS 2025 **(Outstanding Poster Presentation Award)**
3. Yongkee Lee, **Woohyeok Kim**, Sunghyun Cho, "Non-uniform Exposure Prediction Pipeline for Burst Image Restoration", KCGS 2025
4. Daeyeon Kim, **Woohyeok Kim**, Sunghyun Cho, "Enhancing A Neural-Network-based ISP Model through Positional Encoding", KCGS 2024 **(Undergraduate Outstanding Paper Award)**
5. **Woohyeok Kim**, Geonu Kim, Seungyong Lee, Seung-Hwan Baek, Sunghyun Cho, "Improvement of Realistic Forward and Reverse ISPs through Modular ISP Networks", IPIU 2024 **(Outstanding Paper Award)**

MILITARY SERVICE

- Republic of Korea Army (2nd Infantry Division Reconnaissance Battalion), Yanggu, Gangwon-do, Korea (Feb. 2018 ~ Oct. 2019) **(Chief of Staff of the Army Award)**

AWARDS & HONORS

- Industry-Academia Outstanding Paper Award, Samsung Electronics, Korea (Aug. 2026)
- Top 22 Team, GARL Challenge 2026, Korea (Jul. 2026)
- Outstanding Reviewer, NeurIPS 2025, United States (Dec. 2025)
- Outstanding Poster Presentation Award, KCGS 2025, Korea (Jul. 2025)
- Undergraduate Outstanding Paper Award, KCGS 2024, Korea (Jul. 2024)
- Outstanding Paper Award (Bronze Prize), IPIU 2024, Korea (Jan. 2024)
- Chief of Staff of the Army Award, Republic of Korea Army, Korea (Nov. 2018)

PROJECTS

- Development of an End-to-End Learning Framework from Camera Capture Settings to ISP, Samsung Electronics (DS Division), Korea (Oct. 2025 ~ Present)
- Augmentation Technologies to Overcome the Limits and Disabilities of Human Cognition and Intelligence, ITRC/MSIT, Korea (Jul. 2024 ~ Present)
- Development of Neural Camera ISPs for Multi-Degradation Restoration and Perceptual Image Enhancement, MSIT, Korea (Mar. 2023 ~ Feb. 2026)
- End-to-End Wide-Baseline Enhancement via Robust Alignment from Raw Image Bursts with Multiple Degradations, SAIT, Korea (Nov. 2022 ~ Oct. 2023)
- High-Quality Image Restoration Reflecting Real Camera Characteristics, Samsung Electronics (MX Division), Korea (Aug. 2022 ~ Jul. 2023)

PATENTS

1. Sunghyun Cho, Jaesung Rim, **Woohyeok Kim**, Heemin Yang, "Server and Method for Providing a Video Correction Service", 10-2026-0086973 (KR, issued), TBD (US, issued)
2. Sunghyun Cho, **Woohyeok Kim**, Jaesung Rim, Daeyeon Kim, "Method for Restoring Burst Image and Device the same", 10-2025-0080965 (KR, issued), 19/547,574 (US, issued)
3. Minsu Cho, Sanghyun Kim, Minjung Lee, Deunsol Jung, Manjin Kim, **Woohyeok Kim**, Jaesung Rim, Sunghyun Cho, "Base Frame Selection for Burst Image Enhancement on Multiple ExposureTime Images", 10-2023-0154857 (KR, issued), 18/934,024 (US, issued)

ADDITIONAL INFORMATION

- Expected Graduation: August 2028
- Programming Languages: Python, C/C++, MATLAB